

Sanctioned loan amount prediction

FIT5149 Assessment 1

Group 43

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1. Problem and main result

We predict the sanctioned amount in AUD for 10,000 unseen loan applications. Our final model is two-stage CatBoost: it estimates approval probability and the likely approved share separately. It achieved mean five-fold RMSE of AUD 33,958.77 and public Kaggle RMSE of AUD 33,096.96. We selected it using internal validation; the private Kaggle score is still unknown.

2. What we found in the data

The training set has 19,322 rows and 29 columns, including the target. The test set has 28 input columns. We found no duplicate rows or IDs, and no IDs shared between the sets. All labelled sanctioned amounts are between zero and the requested amount.

There are 5,177 zero-sanction cases, or 26.79% of training rows. Rounded sanctioned-to-requested ratios take eight values: 0%, 65%, 70%, 75%, 80%, 85%, 90% and 100%. This pattern suggested modelling the lending decision and the approved share rather than only the dollar amount.

Raw missing values	Train	Test
property_age_years	32.40%	32.62%
monthly_income_aud	25.69%	24.84%
annual_income_aud	25.09%	24.85%
employment_sector	24.36%	24.19%
credit_rating	20.46%	21.33%

The missingness rates are similar across train and test, but this does not mean gaps are random or harmless. We kept missing-value flags so the model can distinguish an unavailable value from an observed one.

We also found 406 occurrences of -999: 100 in repayments, 107 in co-applicant status and 199 in property value. These codes were changed to missing. Property age matched annual income exactly in all 11,524 overlapping rows and had no values from 0 to 200, so we dropped its numeric values rather than interpret them as ages.

3. Cleaning and feature choices

We apply the same row-by-row cleaning to training and test data. Application ID is kept only for the submission. We also exclude property_ref, an internal reference, and the corrupted property-age values. The final matrix contains 49 predictors, including 12 categorical inputs.

Annual and monthly income are put on the same annual scale and combined using the median of the available values within each row. We retain the original fields, count the available income sources, and record their relative disagreement. Loan-to-income, loan-to-property-value and repayment-to-monthly-income ratios describe affordability and security; property value minus requested amount measures the remaining property value.

Dates are represented by year and sine/cosine terms for month and day of year, keeping the end and start of the calendar close together. CatBoost handles numeric missing values directly. Missing categorical values receive a separate string label. In the earlier Ridge, Random Forest and HGB comparisons, imputation and encoding were fitted inside each training fold; Ridge also used scaling.

We checked whether selected preprocessing decisions helped, instead of assuming that more features would improve the score. These are controlled comparisons within the stated model, not estimates of each feature's value in every model.

Controlled comparison	Mean fold RMSE in AUD
HGB with cleaned raw features	36,111
HGB with ratios and missing flags	36,065
HGB also using property age as an income backup	36,074
Random Forest with / without <code>property_ref</code>	35,994 / 35,956

Adding ratios and missing flags reduced HGB RMSE by about AUD 46. Using property age as an income backup made the result slightly worse. Removing `property_ref` reduced Random Forest RMSE by about AUD 38. These differences are small, so we kept the simpler feature choices without treating the changes as strong evidence of better performance.

4. Validation and error measures

We use five shuffled StratifiedKFold splits with seed 42, balanced by the observed sanction-rate groups. Both stages train only on the fitting rows of each fold. The true sanction rate is never an input feature. Every application receives a held-out prediction, and the same folds are reused across candidates.

RMSE in AUD is the selection metric because it is used by Kaggle and penalises large dollar mistakes. MAE describes the average absolute error and provides another view. MAPE would be unsuitable with 26.79% zero targets. We report mean fold RMSE; the final global OOF RMSE, calculated over all held-out predictions together, is AUD 33,975.51. Repeated model comparisons can make development scores optimistic.

5. Models and final selection

We began with Ridge, Random Forest and Histogram Gradient Boosting (HGB), covering a linear model, bagged trees and boosted trees. We then tested sanction-rate prediction and CatBoost. The table separates the initial direct-amount models from the later target-aware models; all use the original five folds.

Model or formulation	Mean fold RMSE in AUD
Ridge with prediction bounds	43,306
Random Forest after excluding <code>property_ref</code>	35,956
HGB predicting the sanction rate	34,812
HGB using expected sanction-tier probabilities	34,658
Single CatBoost sanction-rate regressor	34,099.76
Two-stage CatBoost selected for attempt 4	33,958.77
Equal average of three two-stage models	33,989.17

For the selected model, a CatBoost classifier learns whether the amount is positive. A second CatBoost regressor learns the rounded sanctioned-to-requested ratio on approved fitting rows only. We multiply approval probability, predicted positive rate and requested amount. Rates and amounts are bounded to their valid ranges. Keeping probabilities avoids forcing an uncertain case into a hard approve/reject decision.

The classifier uses Logloss, depth 5 and 1,200 iterations. The rate regressor uses RMSE, depth 4 and 800 iterations. Both use learning rate 0.03, L2 regularisation 10, random seed 42 and six CPU threads. The objectives follow the CatBoost documentation [1, 2].

Partition seed	Single CatBoost	Two-stage CatBoost
42	34,099.76	33,958.77
2026	34,323.40	34,119.05

Two-stage CatBoost improved all five paired folds in both partitions, lowering mean RMSE by AUD 141 and AUD 204 respectively. Its fold standard deviation on the original split was AUD 1,194. These are modest gains, not proof of significance. The second partition reuses the same applications and is only a stability check.

A bounded comparison of 15 later configurations also tested deeper trees, more iterations, request-based weights, smaller feature sets, direct-amount CatBoost, tier classification and XGBoost. None beat the selected model on the original folds. Two fixed 50/50 blends also failed to improve global OOF RMSE. The last three-seed average was AUD 30 worse on split 42 and AUD 38 better on split 2026, effectively a tie. We retained the simpler two-stage model; no blend weights were fitted to the public leaderboard.

6. Feature influence and model errors

We considered both relationships in the data and changes in held-out prediction error. Pearson correlations with sanctioned amount were 0.740 for requested amount, 0.375 for credit rating and 0.269 for co-applicant status. These are associations, not causal effects, and dollar relationships can partly reflect the size of the loan request.

For the final model, we shuffled five inputs highlighted by the EDA, one at a time in validation rows, without refitting. We repeated each shuffle three times per fold. We averaged the three shuffles within each fold, then calculated the mean and standard deviation across the five folds. This standard deviation is not a confidence interval.

Input shuffled	RMSE increase AUD	Fold SD AUD
Requested amount	51,199	1,842
Credit rating	20,198	1,873
Co-applicant status	17,953	904
Income consistency	1,356	489
Applicant age	60	41

Credit rating and co-applicant status matter for estimating the lending decision, while requested amount sets its dollar scale. Shuffling requested amount also changes the final conversion to AUD, so its effect describes the whole prediction process. Income consistency has a smaller effect and age very little in this model. These checks cover only five inputs; correlated features may share information, so they are not a complete ranking or a causal test.

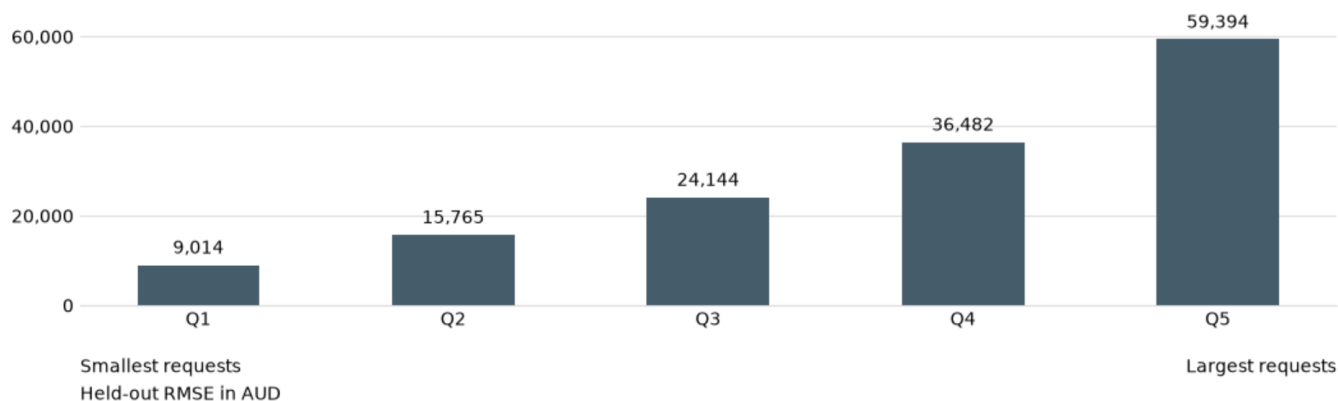


Figure 1. Final-model OOF error by requested-amount quintile. Each group has 3,864 or 3,865 rows.

Errors rise from AUD 9,014 in the smallest request group to AUD 59,394 in the largest. Rejected applications have RMSE of AUD 58,765 ($n = 5,177$), versus AUD 17,690 for approved applications ($n = 14,145$). Missing credit ratings also mark a harder group: RMSE is AUD 40,738 ($n = 3,953$), compared with AUD 32,006 when ratings are available ($n = 15,369$). These diagnostics use the final model, not the earlier Random Forest.

7. Kaggle results and limitations

Attempt	Model	Public RMSE
1	Random Forest	35,514.14046
2	Single CatBoost rate model	33,510.57914
3	Same formulation in clean implementation	33,510.57914
4	Two-stage CatBoost	33,096.96156
5	Three-seed two-stage average	33,119.70974

Attempt 4 improved public RMSE by AUD 413.62 over the earlier CatBoost and was in position 19 in our screenshot. Attempt 5 was AUD 22.75 worse. Positions can change. The public score uses only 2,000 test rows; the private score on the other 8,000 determines the competition marks and was unknown when this report was written.

Attempt 4's public RMSE is AUD 861.81 below its mean fold RMSE. Different samples can produce different errors, especially when a few large loans dominate RMSE. We do not treat this gap as evidence of a further improvement. The model was chosen from cross-validation before attempt 4 was uploaded. The historical attempt 1 CV log and the current Random Forest rerun both round to AUD 35,956.

Limits of the final model

The final OOF MAE is AUD 16,850.52, much lower than RMSE because large mistakes receive extra weight under squared error. If we prioritised MAE, we might choose a different model. For example, the earlier hard-tier HGB had MAE around AUD 14,425 but RMSE around AUD 39,459. A lower typical absolute error did not make it better for the competition metric.

The model still struggles with rejection uncertainty, large loans and missing credit ratings. The 85%, 90% and 100% tiers have only 42, 50 and 40 examples, so their behaviour is poorly supported. We would want better input data, temporal and external validation, and checks across applicant groups before considering any real use. Our findings are limited to the supplied course dataset. The model should not be treated as a causal measure of creditworthiness or a ready lending system.

Reproducibility and conclusion

The standalone notebook FIT5149_A1_final.ipynb reproduced submission.csv exactly, with 10,000 valid rows. FIT5149_A1_final_clean.ipynb brings together the calculations behind the report's tables and Figure 1, including the model comparisons and final diagnostics. Kaggle scores are recorded from our submissions, since the test targets are hidden. We used Python 3.11.15, pandas 3.0.5, NumPy 2.4.6, scikit-learn 1.9.0 and CatBoost 1.2.10. We kept attempt 4 because it improved on the original CatBoost in both validation partitions, while the three-seed average did not show a clear gain.

Method references

- [1] CatBoost documentation. Classification objectives, Logloss. <https://catboost.ai/docs/en/concepts/loss-functions-classification>
- [2] CatBoost documentation. Regression objectives, RMSE. <https://catboost.ai/docs/en/concepts/loss-functions-regression>

Project plan and work division

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Group 43: TEAM ERIC-ADITYA

We agreed on an equal overall contribution of 50% each. We worked on different parts of the assignment and shared our progress through GitHub and WhatsApp.

Group member	Student ID	Contribution
Aditya Palve	35324074	50%
Eric Andres Acosta Ramirez	32642083	50%

Work division

Eric worked on the EDA, missing and invalid values, preprocessing, and the first Random Forest submission. He also organised the repository and worked on the later two-stage CatBoost and three-seed experiments, prediction checks, and the updated report, with AI assistance.

Aditya developed the initial CatBoost approach and prepared the notebooks used for attempts 2 and 3. He also prepared the first PDF report and shared his model results and suggestions for further experiments.

We discussed the results and next steps together. The 50/50 split reflects our agreed overall contribution, rather than the number of commits or Kaggle uploads made by each member.

Project plan and progress

Data preparation: Inspect the data, identify missing or invalid values, and agree on the cleaning steps. The findings are recorded in the analysis notebook.

Model development: Compare initial models, test CatBoost and later variations, and record validation and Kaggle results. The selected implementation reproduces attempt 4.

Reporting: Bring together the EDA findings, model comparisons and final diagnostics. The updated report is ready for both members to review.

Final checks before submission: Review this work statement, sign the cover sheet, complete the AI documents, check the nominated Kaggle entry, and verify the report and ZIP before uploading them.

Use of AI tools

AI assistance was used to review code, explore modelling ideas, and draft and revise parts of the report. The separate AI declaration and chat history document accurately describe how AI was used throughout the work.